# GreenDash: green customer insights you can check

Megathon X · Problem Statement 1 (Lloyds) · Final report. All data is synthetic.

## 1. Solution

GreenDash finds green-finance opportunities for each customer (EV Finance, Green Home Loan, Solar Finance, Paperless Banking), estimates the impact and explains each one to two audiences: the customer and the bank's relationship teams. Every recommendation is a **Proof Card** with three verdicts that are never blended (likelihood from ML, eligibility from rules, impact from a carbon engine), the evidence with dates, an impact range with its formula, and an explanation that is verified before it is shown. When a recommendation cannot be justified, the system abstains, asks one question, or rules the product out and shows the rule.

**Core rule, enforced in code:** ML predicts. Rules decide eligibility. The carbon engine calculates. The LLM only explains. Claim-Check verifies.

## 2. Approach

| Stage | What it does | Key design choice |
|---|---|---|
| Data | 5,000 synthetic customers in the brief's 8 tables, seed 42, cut-off 30 Jun 2026, 90-day outcome window. Electricity levels and seasonality are taken from the provided London smart-meter data (5,364 households, aggregates only). | Take-up labels come from hidden traits plus noise, not from a rule on visible fields. Age, income and postcode are never model inputs. |
| Categoriser (FR2) | Ordered keyword rules, then a character n-gram text model; exactly one of 8 categories. | Boundary decisions (Fuel vs Automotive, Gas/Electricity vs Utilities) live in config. |
| Propensity (FR3) | One calibrated LightGBM model per product; outputs likelihood, a separate confidence score with four parts, and fixed reason codes. | Split by customer (3,494 / 749 / 749) with a leakage test; two baselines; isotonic calibration. |
| Eligibility | Product rules in config (e.g. owns an EV, renter, existing solar, needs paper statements). | Propensity never establishes eligibility. |
| Carbon engine (FR4) | Deterministic formulas using UK Government 2024 conversion factors; returns low / central / high, formula, baseline, horizon and a quality band. | Refuses to estimate when an input is missing. Paperless is reported in sheets, never as carbon. |
| Orchestrator | Assigns one of five statuses: Recommend, Soft surface, Abstain, Review, Ruled out. | It never changes a verdict. |
| Explanation (FR5) | Gemini rewrites a minimised evidence pack; input and output guards; **Claim-Check** verifies every sentence (numbers, cited fields, banned promises, metric family, status). | Any failure falls back to fixed wording. Every result goes to an append-only, hash-chained audit table. |
| Delivery | FastAPI with signed short-lived tokens and role checks; Streamlit app with Customer Hub, business Cockpit (FR6) and a Receipts page. | One command builds everything; the manifest hash repeats across runs. |

## 3. Features

- **Proof Card (FR7):** status, why-now trigger, impact range, three verdicts, evidence with dates and provenance, formula and factor source, one clear action, and “not a credit decision”.
- **Scenario Lab:** seven live changes to a persona (set vehicle to EV, remove mileage, remove energy data, add a conflicting signal, make data stale, mark as renter, sell the car). The pipeline recomputes for that customer and shows before, after, the rule that caused the change, and an invariance check. A 35-case automated suite covers the four scenario types named in the brief.
- **Claim-Check red-team button:** inject “Switching will save you 4,000 kg a year and you’re pre-approved” and watch it blocked for an unsupported number and two banned promises.
- **Business Cockpit:** CO2e, energy, paper and digital adoption reported separately; funnel; propensity × impact quadrant; segments and regions; drill-down Customer → Opportunity → Evidence → Impact → Action; fairness tables and audit-chain status.
- **Receipts page:** held-out metrics against baselines, calibration, categoriser score, explanation checks and scenario results, including the cases where our model does not win.
- **Security:** role-based access, customers cannot read each other, rate limiting, parameterised SQL, pseudonymous keys, encrypted feedback notes, hash-checked models, escaped output, no secrets in the repository.

## 4. Results (held-out test set, 749 customers)

| Product | LightGBM AUC | Logistic regression AUC | Rule AUC | Lift at top 10% | Brier | Base rate |
|---|---|---|---|---|---|---|
| EV Finance | 0.706 | 0.701 | 0.500 | 2.4× | 0.042 | 4.5% |
| Green Home Loan | 0.766 | 0.771 | 0.638 | 4.4× | 0.030 | 3.1% |
| Solar Finance | 0.673 | 0.705 | 0.790 | 2.5× | 0.016 | 1.6% |
| Paperless Banking | 0.895 | 0.882 | 0.874 | 3.4× | 0.047 | 6.0% |

- **Honest reading:** LightGBM is level with logistic regression and behind the simple rule on Solar. Take-up is rare (2–6%), so each test set holds a few dozen positives and differences of a few points are within noise. Scores are calibrated, so a customer is recommended when their likelihood is at least the product's average take-up rate and confidence is at least 0.6.
- **Categoriser:** macro-F1 1.00 against 0.60 for rules alone, 5-fold test on unseen merchants (synthetic merchant names are regular, so real statements will be harder).
- **Scenario suite:** 35 of 35 cases pass.
- **Explanations:** 80 real evidence packs were tested. 55 were written by the model and all 55 passed Claim-Check; the other 25 were refusals, which are explained by fixed rule wording by design. A second model acting as judge rated 78 of 80 (97.5%) faithful to the evidence. Canary leakage was 0%.
- **Adversarial sentences:** 15 of 20 hand-written unsafe sentences are blocked by the sentence rules alone.
- **Engineering:** 549 automated tests pass. The full pipeline runs in about 3 minutes from one command, and two consecutive runs give the same manifest hash (sha256:30c8997d9f5646fb…).

## 5. Scoring criteria

| Criterion | Evidence |
|---|---|
| Business relevance (15%) | One engine, two audiences; one clear action per card; why-now triggers; a ranked, explainable pipeline for relationship teams. |
| ML and recommendation quality (20%) | Customer-level split with leakage test, two baselines, calibration, held-out metrics, labels from hidden traits, honest reporting. |
| Sustainability insight (15%) | Published factors, formula and baseline on every estimate, ranges with a quality band, four metric families kept apart, abstention. |
| GenAI and LLM (15%) | Minimised input, guards, canary, five-rule Claim-Check, fallback, audit trail, evaluation scored by rules and by a second model. |
| Technical architecture (15%) | Frozen contracts, reproducible one-command pipeline, authenticated API, cached explanations, sub-second scenario recompute. |
| Explainability and Responsible AI (10%) | Separate verdicts, reason codes with provenance, rule text for refusals, fairness tables, human review path. |
| UX and innovation (5% + 5%) | Brand design system with light and dark mode; Scenario Lab, Claim-Check and the Receipts page. |

## 6. Limitations and next steps

- Synthetic labels: the metrics show a sound pipeline, not real-world accuracy. The confidence score is not yet shown to track error.
- Sign-in is a role picker with short-lived tokens and storage is a local SQLite file; a bank deployment would use its single sign-on with multi-factor authentication and a managed, encrypted database.
- Parts of the Cockpit (cohort quadrant, named segments, trend chart) are illustrative. The optional customer-memory extension is not built: feedback and opt-outs are stored but do not yet change later recommendations.
- Next: uplift modelling on real outcomes, estimated-versus-realised impact after take-up, and the two optional products added by configuration.